



Tests Documentation: SwayLayoutManager

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Software Libre (LIDSoL)

Areas:

Frontend

Backend

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1 Integration Tests

1.1 CLI Tests (`cli_test.go`)

`TestHelpFlag`

This test validates help flag functionality in the compiled binary and usage information display.

- **Tests:**
 - `long help flag` - Tests `-help` flag behavior and output
 - `short help flag` - Tests `-h` flag behavior and output

`TestVersionFlag`

This test validates version flag functionality in the compiled binary and version information display.

- **Tests:**
 - `long version flag` - Tests `-version` flag output
 - `short version flag` - Tests `-v` flag output

`TestInvalidCommand`

This test validates error handling for invalid CLI commands and graceful failure handling.

- **Tests:**
 - `invalid command` - Tests response to unrecognized commands
 - `typo in command` - Tests response to misspelled commands
 - `empty args` - Tests response when no arguments provided

`TestListCommandEmpty`

This test validates list command behavior when no presets exist and empty state handling.

- **Tests:** List command with empty preset directory shows appropriate message

`TestSaveCommandInvalidName`

This test validates save command robustness with various invalid input scenarios.

- **Tests:**
 - `empty name` - Tests save with empty preset name
 - `whitespace only` - Tests save with whitespace-only names
 - `special characters` - Tests save with special characters in names

TestDeleteCommandNonExistent

This test validates delete command error handling for missing presets.

- **Tests:** Delete command behavior with non-existent preset names

TestBinaryExecution

This test validates binary execution characteristics and process management.

- **Tests:** Binary behavior, performance, and proper execution

1.2 Configuration Tests (config_test.go)

TestXDGEnvironmentVariables

This test validates XDG Base Directory Specification compliance in real environments and Linux desktop standards.

- **Tests:**
 - `respects_XDG_CONFIG_HOME` - Uses custom config directories when `XDG_CONFIG_HOME` set
 - `respects_XDG_DATA_HOME` - Uses custom data directories when `XDG_DATA_HOME` set
 - `falls_back_to_home_directory` - Falls back to `~/.config` and `~/.local/share` appropriately

TestConfigurationPersistence

This test validates configuration loading and directory creation in real environments.

- **Tests:**
 - `creates_directories_correctly` - Creates proper directory structure with correct paths
 - `succeeds_without_existing_files` - Works without pre-existing configuration files

TestDefaultConfiguration

This test validates default configuration behavior and sensible defaults.

- **Tests:**
 - `provides_default_values` - Sets appropriate default values for all config fields
 - `creates_necessary_directories` - Creates required directories with correct permissions (0755)

TestConfigurationDirectoryStructure

This test validates complete application directory structure creation with proper hierarchy.

- **Tests:**
 - `creates_complete_directory_structure` - Creates full XDG directory hierarchy
 - `directory_permissions_are_correct` - Sets proper directory permissions

TestConfigurationIsolation

This test validates configuration isolation between different environments and multi-user scenarios.

- **Tests:**
 - `multiple_configs_with_different_environments` - Ensures different configs don't interfere with each other

1.3 Filesystem Tests (filesystem_test.go)

TestXDGDirectoryCreation

This test validates real XDG directory creation and structure validation with Linux desktop directory standards.

- **Tests:** XDG Base Directory structure creation with proper hierarchy and file persistence

TestPresetFilePersistence

This test validates complete preset save/load cycles with real files, end-to-end preset persistence.

- **Tests:** Complex preset data persistence through complete save/load/verify cycles with multiple presets

TestFilePermissions

This test validates Unix file permission handling for security and access control.

- **Tests:** File permissions (0644) and directory permissions (0755) validation for proper multi-user access

TestJSONFormatValidation

This test validates JSON format validation and complex data structure handling.

- **Tests:** Complex JSON preset validation with full workspace layouts, verifies valid JSON formatting

TestConcurrentAccess

This test validates concurrent file operations and thread safety.

- **Tests:** 5 simultaneous preset save operations to test concurrency handling and prevent race conditions

TestDeleteFileRemoval

This test validates complete file deletion and cleanup operations.

- **Tests:** File deletion with verification of complete removal from filesystem

1.4 Operations/Filtering Tests (operations_test.go)

TestSaveWithWorkspaceFiltering

This test validates save operation with workspace filtering applied.

- **Tests:**
 - `save and skip specific workspace` - Saves layout excluding specified workspaces
 - `save only specific workspace` - Saves layout including only specified workspaces

TestSaveWithAppFiltering

This test validates save operation with application-level filtering.

- **Tests:**
 - `save and skip specific app` - Saves layout excluding specific applications
 - `save only specific app` - Saves layout including only specific applications

TestSaveWithFloatingFilter

This test validates save operation excluding floating windows.

- **Tests:**
 - `filter floating windows` - Saves layout without floating containers

TestListWithFilteredPresets

This test validates list operation with preset metadata validation.

- **Tests:** Lists presets and verifies metadata contains correct information

TestDeleteFilteredPresets

This test validates delete operation with verification of file removal.

- **Tests:** Deletes preset and confirms complete removal from filesystem

1.5 Restoration Tests (restore_test.go)

TestRestoreBasicFunctionality

This test validates core restore operation functionality with empty and simple presets.

- **Tests:**

- `restore empty preset` - Handles preset with no workspaces, verifies zero counters

TestRestoreWithReuseConfiguration

This test validates restore operation with window reuse settings enabled.

- **Tests:**
 - `configure reuse existing windows` - Tests `ReuseExistingWindows` setting in restore operation
 - `configure reuse specific apps` - Tests `ReuseApps` array in restore operation

TestRestoreWithClearConfiguration

This test validates restore operation with workspace clearing settings.

- **Tests:**
 - `configure clear workspaces` - Tests `ClearWorkspaces` setting in restore operation
 - `configure clear all workspaces` - Tests `ClearAllWorkspaces` setting in restore operation

TestRestoreErrorHandling

This test validates restore operation error handling with invalid inputs.

- **Tests:**
 - `restore nil preset` - Handles nil preset parameter with appropriate error
 - `restore preset with invalid workspace` - Handles presets with invalid workspace data

TestRestoreResultTracking

This test validates result tracking and statistics for restore operations.

- **Tests:**
 - `result tracks workspaces` - Verifies workspace restoration counters
 - `result structure is complete` - Validates all result fields are properly populated

1.6 Sway Environment Tests (`sway_test.go`)

TestSwayAvailability

This test validates Sway window manager detection and availability with graceful degradation.

- **Tests:**
 - `swaymsg command availability` - Tests `swaymsg` detection in `PATH`

- `sway client creation without swaymsg` - Tests failure handling when Sway unavailable

TestSwayConnectionAttempts

This test validates live connection to running Sway instance with real Sway v1.11.0 integration.

- **Tests:**
 - `get sway version` - Retrieves live Sway version information
 - `get sway tree` - Fetches complete workspace tree from live environment
 - `get sway workspaces` - Gets workspace list from running Sway

TestSwayJSONParsing

This test validates JSON parsing of complex Sway data structures and Sway IPC response handling.

- **Tests:**
 - `parse sway version JSON` - Version information JSON parsing and validation
 - `parse sway workspace JSON` - Workspace data structure JSON parsing
 - `parse complex sway tree JSON` - Complex tree with `idle_inhibitors` and nested structures

TestSwayCommandTimeout

This test validates Sway command execution performance and timeout handling.

- **Tests:**
 - `version command completes quickly` - Version command performance validation (< 10 seconds)
 - `tree command completes quickly` - Tree command performance validation

TestSwayErrorHandling

This test validates error handling and degradation for Sway operations.

- **Tests:**
 - `handles sway not running` - Graceful handling when Sway is not available

2 Unit Tests

2.1 Configuration Tests (`config_test.go`)

TestNewConfig

This test validates configuration constructor functionality and XDG Base Directory specification compliance.

- **Tests:**

- **with default XDG paths** - Creates config using default `~/.local/share` and `~/.config` paths
- **with custom XDG paths** - Uses custom `XDG_DATA_HOME` and `XDG_CONFIG_HOME` when set

TestConfigPresetPath

This test validates preset file path generation functionality and file path construction with JSON extension handling.

- **Tests:**

- **name without extension** - Adds `.json` extension to preset names without extension
- **name with `.json` extension** - Handles preset names that already have `.json` extension
- **name with other extension** - Appends `.json` to names with non-`.json` extensions
- **empty name** - Handles empty preset names

TestConfigDefaultPresetName

This test validates default preset name generation with PID-based naming.

- **Tests:** Validates format “`layout-<pid>`” and ensures PID is present

2.2 Layout Processing Tests (`layout_test.go`)

TestLayoutTypes

This test validates layout data structure definitions and JSON serialization for data structures.

- **Tests:**

- **Preset JSON marshaling** - Complete Preset struct JSON marshal/unmarshal cycle
- **OutputLayout JSON marshaling** - OutputLayout struct JSON compatibility
- **WorkspaceLayout JSON marshaling** - WorkspaceLayout struct serialization
- **ContainerLayout with ExecCommand JSON marshaling** - Container with exec commands

TestNewParser

This test validates layout parser initialization with Sway client dependency.

- **Tests:** Creates parser with valid Sway client or skips if unavailable

TestCaptureCurrentLayout

This test validates live workspace layout capture from Sway and real-time layout data extraction.

- **Tests:**
 - `successful layout capture attempt` - Captures layout from running Sway instance or handles non-Sway environments gracefully

2.3 Operations/Filtering Tests (operations_test.go)

TestFilterSkipWorkspaces

This test validates workspace filtering by skip list with numbers and names.

- **Tests:**
 - `skip by number` - Filters workspaces by numeric identifier
 - `skip by name` - Filters workspaces by workspace name
 - `skip multiple workspaces` - Removes multiple specified workspaces
 - `skip non-existent workspace` - Handles skipping workspaces that don't exist
 - `empty skip list` - No filtering when skip list is empty

TestFilterOnlyWorkspaces

This test validates workspace filtering to keep only specified workspaces.

- **Tests:**
 - `only by number` - Keeps only workspaces matching number
 - `only by name` - Keeps only workspaces matching name
 - `only multiple workspaces` - Keeps multiple specified workspaces
 - `empty only list` - Returns empty when only list is empty

TestFilterFloatingContainers

This test validates filtering to exclude floating windows from layout.

- **Tests:** Removes containers with floating type or floating enabled

TestFilterSkipApps

This test validates application filtering by skip list using `app_id` and window class.

- **Tests:**
 - `skip by app_id` - Filters containers by application ID
 - `skip by class` - Filters containers by window class
 - `skip multiple apps` - Removes multiple specified applications

- `empty skip list` - No filtering when skip list is empty

TestFilterOnlyApps

This test validates application filtering to keep only specified applications.

- **Tests:**
 - only by `app_id` - Keeps only containers matching app ID
 - only by `class` - Keeps only containers matching window class
 - Additional subtests for multiple apps and empty list

TestFilterNestedContainers

This test validates recursive filtering within nested container hierarchies.

- **Tests:** Applies filters recursively to nested container structures

2.4 Preset Management Tests (`preset_test.go`)

TestNewManager

This test validates preset manager initialization with configuration dependencies.

- **Tests:** Creates manager with valid config and verifies non-nil return

TestSavePreset

This test validates core preset saving functionality, complete save operation from layout data to file creation.

- **Tests:**
 - `successful save` - Saves valid preset data and creates JSON file
 - `save preset with empty name` - Handles empty preset names with error

TestLoadPreset

This test validates preset loading and validation functionality with file reading, JSON parsing, and data validation.

- **Tests:**
 - `load existing preset` - Loads and validates previously saved preset
 - `load non-existent preset` - Handles missing preset files with error

TestDeletePreset

This test validates preset deletion functionality and file removal with cleanup operations.

- **Tests:**

- `delete existing preset` - Removes preset file and verifies deletion
- `delete non-existent preset` - Handles deletion of missing presets

TestExistsPreset

This test validates preset existence checking functionality without loading content.

- **Tests:** Correctly identifies existing and non-existing presets

TestListPresets

This test validates preset enumeration and listing functionality with directory scanning.

- **Tests:**
 - `empty list` - Handles empty preset directory
 - `list with presets` - Lists multiple saved presets and validates contents

2.5 Restoration Tests (`restorer_test.go`)

TestNewRestorer

This test validates restorer initialization with Sway client dependency.

- **Tests:** Creates restorer with valid Sway client and verifies non-nil return

TestRestorerConfiguration

This test validates restorer configuration options and settings for reuse and clear operations.

- **Tests:**
 - `default configuration` - Validates default settings (all false, empty arrays)
 - `reuse existing windows configuration` - Tests `ReuseExistingWindows` setting
 - `reuse specific apps configuration` - Tests `ReuseApps` array configuration
 - `clear workspaces configuration` - Tests `ClearWorkspaces` setting
 - `clear all workspaces configuration` - Tests `ClearAllWorkspaces` setting

TestRestoreValidation

This test validates restore operation input validation and error handling.

- **Tests:**
 - `restore with nil preset` - Handles nil preset parameter with error
 - `restore with empty preset` - Handles preset with no workspaces gracefully

TestRestoreResultStructure

This test validates RestoreResult data structure for tracking restore operations.

- **Tests:** Validates result structure fields (WorkspacesRestored, ApplicationsLaunched, ApplicationsReused, ApplicationsFailed, Errors)

2.6 Sway Tests (sway_test.go)

TestNewClient

This test validates Sway client creation and initialization with connection setup to Sway window manager.

- **Tests:**
 - `successful client creation when swaymsg exists` - Creates client when swaymsg available in PATH
 - `client creation fails when swaymsg not available` - Handles missing swaymsg gracefully

TestClientMethods

This test validates Sway client method functionality and Sway IPC command execution with response handling.

- **Tests:**
 - `GetTree` method - Retrieves complete workspace tree from Sway
 - `GetWorkspaces` method - Gets list of all workspaces
 - `GetVersion` method - Retrieves Sway version information

TestSwayTypes

This test validates Sway data structure definitions and JSON compatibility for Sway IPC responses.

- **Tests:**
 - `Node JSON unmarshaling` - Node structure JSON parsing and field validation
 - `Workspace JSON unmarshaling` - Workspace structure JSON compatibility
 - `Version JSON unmarshaling` - Version information JSON parsing